AR-Based Intelligent Automobile Repair Learning & Evaluation System

Proposal presentation -i

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Introduction

Purpose

- Improve automobile vocational training using AI and AR technologies by providing intelligent practical evaluation and real-time safety feedback.

Problem

- Traditional automobile training heavily depends on manual supervision, limited vehicle access, and lacks objective assessment methods and safety monitoring.

Solution

- Our proposed AR-based system integrates AI-powered scoring, safety reasoning, and personalized evaluation reports for practical automobile repair learning.

Impact

- Enhances practical learning, reduces safety risks, improves assessment accuracy, and supports scalable vocational education.



Gap

There is no integrated ***AR-based learning system*** capable of:

- Objective practical assessment
- Real-time safety alerting
- Personalized performance analytics
- Intelligent feedback generation



Solution

Our system introduces four intelligent functionalities:

- AI-Based Scoring System
- Safety Rule Reasoning Engine
- ML Performance Analyzer
- Automated Evaluation Reports

Current Progress Overview

Complete

- Literature Review
- Requirement Analysis
- System Architecture Design
- Safety Rule Logic Design
- UI Mockups
- ML Model Training
- Testing & Validation

In-Progress

- AI Scoring Engine Development
- Backend Development
- Dashboard Integration

Upcoming

- Final Integration
- AR Parts Integration

Application of Knowledge

Modular Architecture

Each module is independently scalable and maintainable.

AI & ML Integration

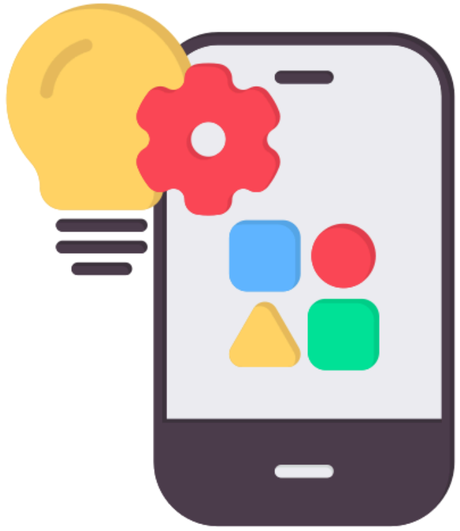
Applied machine learning algorithms for scoring and analytics.

Rule-Based Expert Systems

Used logical safety reasoning mechanisms for hazard detection.

UI / UX Principles

Designed intuitive dashboards and real-time alert interfaces.



Commercialization & Entrepreneurship Potential

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Primary Target Audience / Users

- Vocational training institutes
- Automobile engineering colleges
- Automotive workshops
- Vehicle service centers
- Automotive instructors and trainees



Evidence of Business Potential

- Supports modern vocational education
- Reduces practical training risks
- Scalable for large training institutes
- AI-powered evaluation reduces instructor workload

Functional Requirements

- Real-Time Performance Tracking
- Safety Alert Generation
- Automated Scoring
- Feedback Report Generation
- Safety Rule Management



Technology Stack Used Across Modules

Category	Technologies
Programming	Flutter/Python
Machine Learning	TensorFlow / PyTorch
Computer Vision	OpenCV
AR Development	Unity
Backend	Flask
Database	Firebase
Version Control	GitHub



Vision-Based Engine
Component Identification

Purpose

Identify automobile components using Computer Vision and CNN models.

Functional Progress

- Dataset collection completed
- Image preprocessing completed
- CNN model structure designed
- Initial training completed
- Component detection UI completed

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Vision-Based Engine
Component Identification

Research Application

- Computer Vision
- Convolutional Neural Networks (CNN)
- Object Detection Techniques



Proof of Concept

- Detects automobile components through camera input
- Displays detected component names in real-time

Risks & Mitigation

- ***Risk:*** Poor detection accuracy under low lighting.
- ***Mitigation:*** Applied preprocessing and data augmentation techniques.

Student Task Recognition Based on Identified Component

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Purpose

Recognize repair tasks performed by students using ML classification.

Functional Progress

- Task datasets prepared
- Activity classification logic developed
- Initial ML models implemented

IT22231864 - JAYALATH S.P.S.C.K

Student Task Recognition Based on Identified Component ...

Research Application

- Machine Learning Classification
- Sequential Task Analysis
- Pattern Recognition



Proof of Concept

- Predicts repair tasks from user interactions
- Verifies correct repair sequence

Risks & Mitigation

- ***Risk:*** Incorrect task classification..
- ***Mitigation:*** Increase training datasets and optimize classification models.

AR-Supported Task Verification and Instruction System

Purpose

Provide real-time AR guidance and corrective instructions during repair tasks.

Functional Progress

- Data set training
- Instruction overlays implemented
- Real-time interaction logic developed

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AR-Supported Task Verification and Instruction System

Research Application

- Augmented Reality
- Human Computer Interaction (HCI)
- Interactive Learning Systems



Proof of Concept

- Displays step-by-step repair instructions
- Highlights automobile components using AR overlays

Risks & Mitigation

- ***Risk:*** AR latency during real-time interaction.
- ***Mitigation:*** Optimize rendering and reduce processing load.

Practical Evaluation & Safety Feedback

Purpose

Provide intelligent practical evaluation and real-time safety feedback.

Functional Progress

- AI scoring logic completed
- Safety rule engine designed
- Dashboard UI completed

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Practical Evaluation & Safety Feedback

Research Application

- Artificial Intelligence
- Machine Learning Analytics
- Rule-Based Expert Systems



Proof of Concept

- Calculates performance scores
- Generates Feedback

Risks & Mitigation

- ***Risk:*** Incorrect automated evaluations.
- ***Mitigation:*** Validate AI scores with automotive expert assessments.

Conclusion

Completed Milestones

- Literature Review
- Requirement Analysis
- UI/UX Design
- Initial AI Model Development
- ML Model Training
- Testing & Validation

Next Steps

- Full module integration
- Final AI training
- AR Parts Integration
- User testing with trainees
- Final research documentation




THANK
YOU!